GS FOUNDATION 2024-25

GEOGRAPHY - 6

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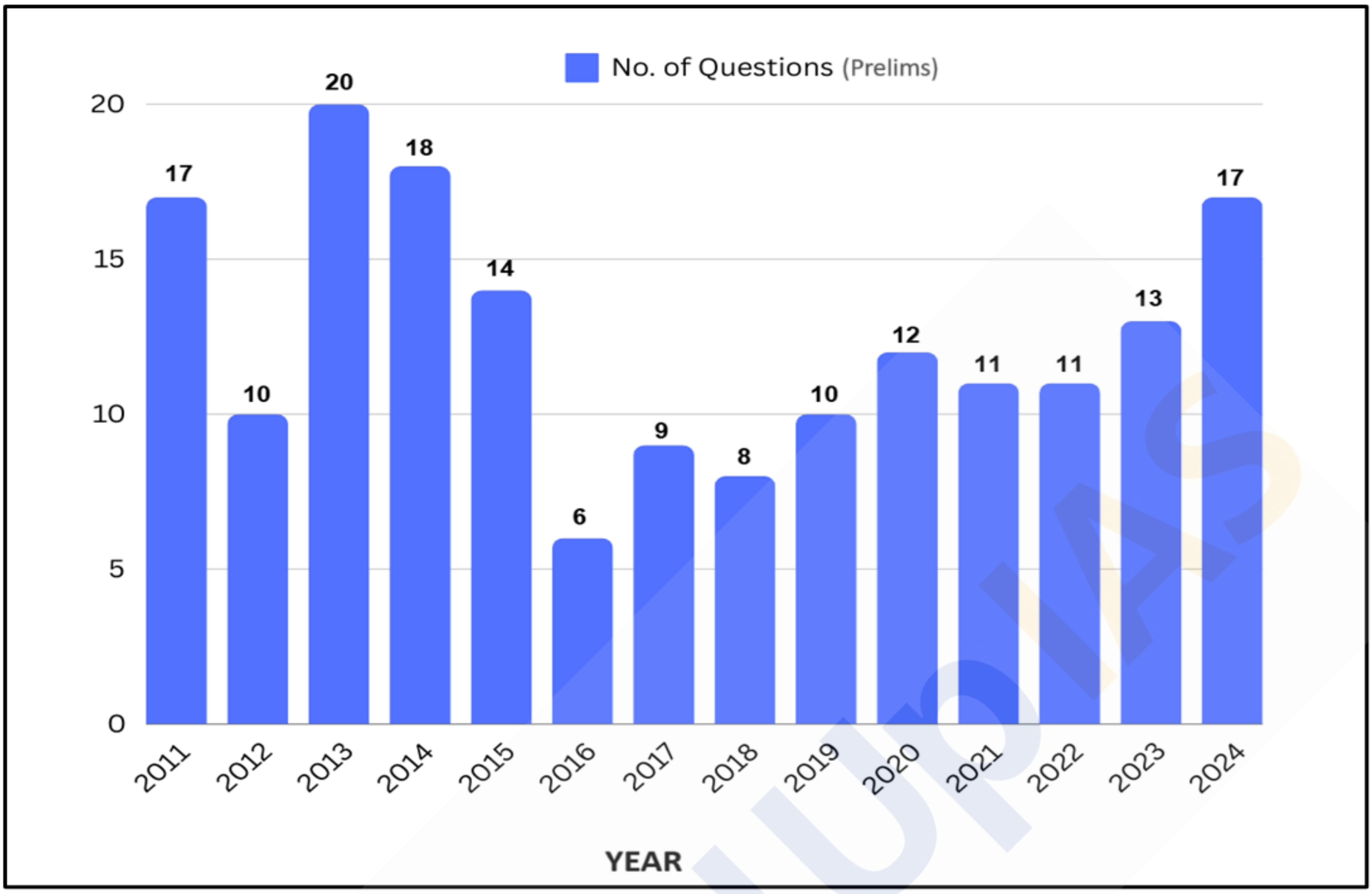

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Air Masses

Introduction:

An air mass is an extensive portion of the atmosphere having **uniform characteristics of temperature, pressure and moisture** which are relatively homogeneous horizontally. Air masses can travel thousands of kilometres in any direction and can reach the stratosphere—16 kilometres (10 miles) into the atmosphere. **Air mass was a concept developed during World War-I by Bjerkens & Bjerknes.** It was founded in the 1930s to predict short-term forecasting (around 24-36 hours).

Key Characteristics of Air Masses:

Temperature and Moisture Uniformity:

- A defining feature of air masses is their relative **uniformity of temperature and moisture content** throughout a horizontal plane at a given altitude. This uniformity arises because air masses develop over vast source regions (like oceans or continents) with similar underlying surfaces for extended periods. The air acquires the thermal properties of the surface and exchanges moisture through evaporation or remains relatively dry over land.

Source Region:

- **The homogenous surfaces** over which air masses form are known as **source regions.** Key source regions include the **high-pressure belts in the subtropics** (tropical air masses) and the **polar regions** (polar air masses). The origin or formation region of an air mass significantly influences its characteristics. Air masses forming over **warm tropical oceans will be warm and moist** (maritime tropical), while those originating over **cold polar regions will be cold and dry** (continental polar). This source region determines the air mass's initial temperature and moisture content, which it carries until it moves or undergoes modification.

Horizontal Scale:

- Air masses are **massive, spanning thousands of kilometers horizontally.** This vast size allows them to maintain their characteristic temperature and moisture properties for a considerable time, even as they move across the globe.

Movement and Modification:

- While air masses are vast, they are not stationary. Winds transport them across the Earth's surface, bringing their weather characteristics to new regions. As air masses **move away from their source regions, they begin to interact with different underlying surfaces. This can modify their temperature and moisture content.** For example, a warm and moist maritime air mass moving over a cooler landmass will gradually cool down on the bottom, leading to increased stability and potentially fog formation.

convergence ← Polar front

Impact on Weather:

Zone b/w Polar Easterly & sub-tropical Westerly.

- The movement and interaction of air masses are fundamental to weather patterns. When **air masses with contrasting temperatures and moisture content meet**, boundaries called **fronts develop.** These fronts separate distinct air masses and are zones of significant weather activity, including precipitation, cloud formation, and wind shear. Understanding air mass movement and fronts is essential for weather forecasting.

Conditions for Formation of Air Masses:

Large Homogeneous Surface:

- The source region for an air mass should be **extensive and relatively uniform.** This means areas like **vast oceans, deserts, or ice-covered regions** are ideal. These large surfaces provide a consistent environment where the air can acquire uniform characteristics over time. For instance, a large ocean surface ensures that the air mass formed over it will have a consistent level of moisture and temperature.

Stable Atmospheric Conditions:

- For an air mass to acquire and retain the properties of the underlying surface, the atmosphere above that surface needs to be stable for an extended period. Stability in the atmosphere **prevents significant mixing of air layers, allowing the air in contact with the surface to adopt its characteristics** without interference from different air masses. This stability helps in maintaining consistent temperature and humidity levels in the developing air mass.

Weak Winds:

- Calm or light wind conditions are crucial for the formation of air masses. **When winds are weak, the air remains in prolonged contact with the surface, allowing sufficient time for the air to reach equilibrium with the underlying surface's temperature and moisture characteristics.** Strong winds would disrupt this process by mixing the air and introducing different characteristics, thus preventing the formation of a uniform air mass.

Modifications of Air Masses:

- Air masses, after originating in their source regions, move away carrying most of their initial physical properties. These properties include:
  - **Lapse rate** of temperature (vertical distribution of temperature)
  - **Mean temperature**
  - **Moisture** content in the air
  - **Wind velocity**
- The **horizontal component** of an air mass more or less maintains **more or less uniformity** in terms of physical properties, whereas the **vertical component shows variations.** As the air mass moves from its source region, it affects the weather conditions of the areas it visits. Simultaneously, the air mass itself undergoes modifications in its physical properties due to the characteristics of the visited area.

Thermodynamic Modifications:

- These involve **heating or cooling of the air mass from below** as it passes over different surfaces away from the source region. Heating of an air mass decreases the vertical stability of the atmosphere. When air masses move out of their source regions to other areas, they modify the weather conditions of those regions and in turn get modified themselves.
- When an **air mass moves over a warmer surface, it heats from below, becomes unstable, and may cause condensation, cloud formation, and precipitation if the air mass contains sufficient moisture.** Conversely, if the air mass is warmer than the surface, it cools from below, stabilizes, and prevents cloud formation. **Cold polar air masses become unstable over warmer surfaces,** while **warm tropical air masses stabilize over colder surfaces.**

Mechanical Modifications:

- Mechanical or dynamic modifications involve **vertical** (upward uplift and downward subsidence) **and horizontal** (advectional) **movement of air,** resulting in changes in the air mass. An air mass is termed stable when the air descends, and unstable when there is upward movement of air. **Mechanical modifications occur due to:**
  - **Turbulent mixing**
  - **Divergence and convergence of air masses**
  - **Subsidence and lateral expansion** (anticyclonic conditions)
  - **Lifting and convergence at the ground surface** (cyclonic conditions)
  - **Advection**

Classification of Air Masses:

The Bergeron classification is the most widely accepted form of air mass classification, though others have produced more refined versions of this scheme over different regions of the globe. Air mass classification involves **three letters.**

The **first letter** describes its **moisture properties:**

"c" represents **continental air masses (dry),**

"m" represents **maritime air masses (moist).**

Its **source region** follows:

"T" stands for **Tropical,**

"P" stands for **Polar,**

"A" stands for **Arctic or Antarctic,**

"M" stands for **monsoon,**

"E" stands for **Equatorial,**

"S" stands for **adiabatically drying and warming air formed by significant downward motion in the atmosphere.**

The **stability of an air mass** is sometimes shown using a **third letter,** either **"k"** (air mass colder than the surface below it) or **"w"** (air mass warmer than the surface below it).

As mentioned above, the air masses are broadly divided into **polar and tropical** air masses. Both the polar and the continental air masses can either have maritime or continental source regions.

There are five major air masses are recognized:

- **Continental Polar (cP)**
- **Maritime Polar (mP)**
- **Continental Tropical (cT)**
- **Maritime Tropical (mT)**
- **Continental Arctic (cA)**

Continental Polar Air Mass (cP):

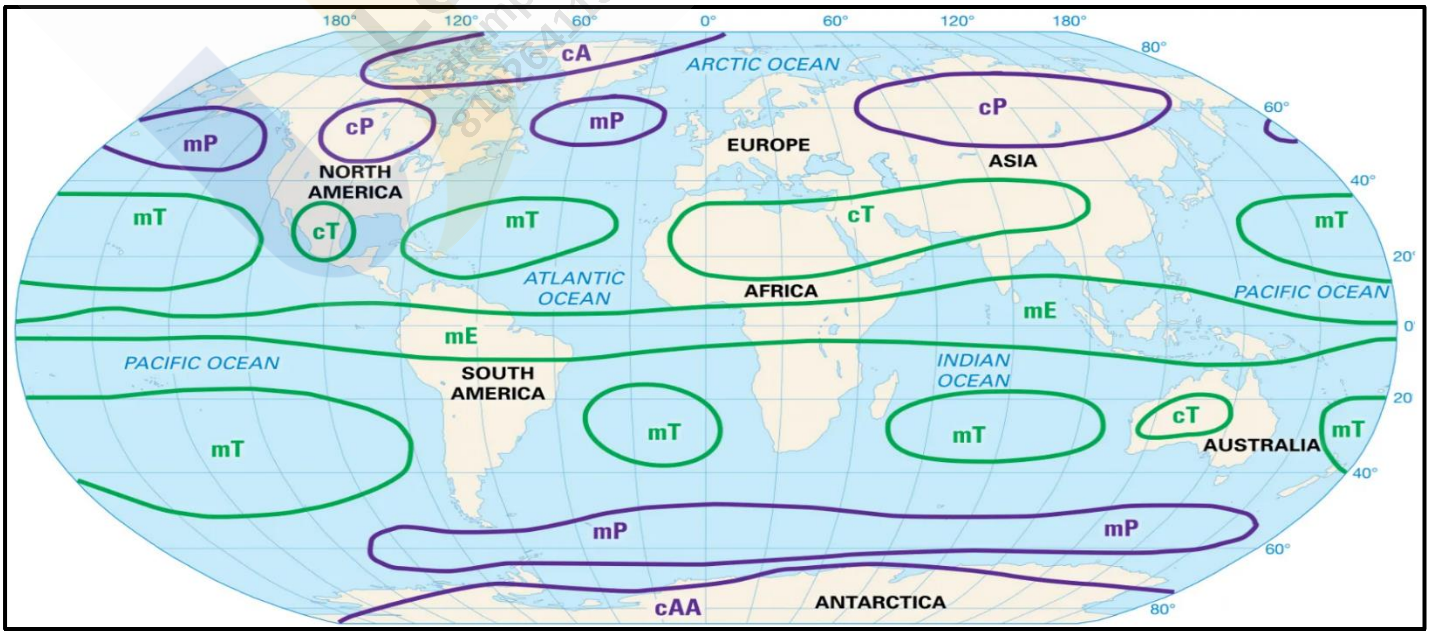
- Polar continental air masses originate over the cold surfaces of **central Canada and Siberia,** moving outward and undergoing thermodynamic and mechanical modifications. These air masses are generally cold and dry. When they **pass over warmer surfaces, they are heated from below, becoming unstable and slightly moist, leading to the formation of low stratocumulus clouds.**
- In winter, the high-latitude source areas are frozen, making the air mass cold, dry, and stable. This results in intense cold waves in regions visited by these air masses, such as the Mississippi plains in the USA. Even in summer, frost conditions can extend as far south as New Orleans, Galveston, and Houston.
- During summer, although the snow-covered surfaces in central Canada and Siberia melt, the polar continental air masses remain cool and dry. Moving over oceanic surfaces warms them from below, making them warmer and occasionally causing **precipitation through cumulus or low stratocumulus clouds.**

Maritime Polar Air Masses (mP):

- Maritime polar air masses (mP) originate from the **same source regions as continental polar air masses (cP).** When cP air masses move over the high-latitude ocean surfaces, they are heated from below by the warmer open ocean waters. This transformation turns them into maritime polar air masses (mP), characterized by a **higher temperature lapse rate and convective instability** in their lower parts. The upper parts remain dry and cool.
- As these modified mP air masses **encounter mountain barriers, they are mechanically lifted, becoming unstable and leading to condensation and significant precipitation on the windward slopes of mountains.** Upon descending on the leeward side of the mountains, they are adiabatically warmed and transform into stable, dry continental air masses. This phenomenon is observed along the west coasts of North America, where the Coast Ranges receive substantial precipitation, while the eastern slopes of the Rockies and the Great Plains remain dry.

Continental Tropical Air Masses (cT):

- Continental tropical air masses originate from **subtropical high-pressure areas,** typically found **in hot deserts between 20° to 30° latitudes** in both hemispheres. These regions are marked by **downward vertical movement** and outward horizontal divergence of winds.
- cT air masses are characterized by **extremely high temperatures,** often exceeding 40°C, very low moisture content, a **steep lapse rate, atmospheric stability, and dry weather conditions.** These air masses generally remain localized in their source regions but, when they do move over ocean surfaces, they undergo modifications and transform into maritime tropical air masses.



Maritime Tropical Air Masses (mT):

- Maritime tropical air masses originate over **warm ocean surfaces within the tropical regions,** typically between **30°N and 30°S latitudes.** These air masses are **warm, moist, and unstable, covering expansive areas.** They are characterized by **convective instability,** leading to the formation of cumulus and cumulonimbus clouds, which often result in abundant rainfall, especially when the air mass encounters frontal activity or is forced to rise over mountain barriers.
- As maritime tropical air masses move poleward, they undergo modifications: over colder ocean waters or land surfaces, they tend to become more stable. Conversely, when these **air masses move over warm land surfaces, they become unstable, intensifying convective processes.**

Continental Arctic Air Masses (cA):

- Continental Arctic (cA) air masses originate over the **extremely cold Arctic regions,** particularly from areas such as the **polar ice caps, northern Canada, Siberia, and Greenland.** These regions, characterized by extensive ice and snow cover, experience minimal solar radiation, especially during the winter months.
- The persistent **lack of sunlight leads to intense surface cooling,** resulting in the formation of cA air masses. These air masses are defined by their frigid temperatures, often plunging well below freezing, and exceptionally low humidity due to the limited moisture content in the cold Arctic atmosphere.
- As cA air masses move southward, they bring their defining cold and dry characteristics with them, significantly impacting the weather patterns of the regions they traverse.
- **The stability of these air masses typically results in clear skies and calm weather, but when they encounter warmer and moister air, the interaction can trigger dramatic weather events, including blizzards and abrupt temperature drops.** The presence of cA air masses is a major factor in the harsh winter conditions experienced in many parts of the Northern Hemisphere.

Climatic Significance of Air Masses:

- **Temperature control -** Air masses act as **giant heat transporters, moving warm or cold air from their source regions to other parts of the globe.** For example, maritime tropical air masses bring warm and humid conditions, while continental polar air masses bring cold and dry weather.
- **Precipitation -** The moisture content of air masses is a key factor in precipitation. **Maritime air masses, originating over oceans, tend to be laden with moisture and can cause rainfall when they move over land.** Conversely, continental air masses are typically drier and contribute to less precipitation.
- **Weather fronts -** When air masses with contrasting temperatures and moisture content meet, they create boundaries called weather fronts. These fronts are **zones of atmospheric instability** where weather phenomena like cyclones, thunderstorms, and precipitation often occur.
- **Regional climates -** The prevailing air masses that dominate a region largely determine its overall climate. For instance, areas influenced by maritime tropical air masses tend to be warm and humid, while those dominated by continental polar air masses experience colder and drier climates.
- **Droughts and heatwaves -** Persistent high-pressure systems associated with certain air masses can lead to **prolonged periods of dry weather and high temperatures, contributing to droughts and heatwaves.**

In essence, air masses act as the building blocks of weather patterns and influence everything from daily fluctuations in temperature and precipitation to the long-term climatic characteristics of a region. Understanding air masses is essential for meteorologists to forecast weather patterns and for climatologists to study climate change.

Fronts

Introduction:

A front is the **sloping boundary between two converging air masses** with **contrasting characteristics** such as temperature, velocity, humidity, and density. It represents a **transitional zone between the weather conditions** of these two air masses. Essentially, a front acts as a line of discontinuity indicating the transition zone between two air masses with distinct weather conditions.

The **process of formation of the fronts** is known as **'frontogenesis.'** It is defined as the process associated with the creation of new fronts or the regeneration of existing, decaying fronts. The region where contrasting air masses converge is known as the region of frontogenesis.

The **process of destruction or dying of fronts** is called **'frontolysis'** which is opposite to the process of frontogenesis. The frontolysis may be affected when either two contrasting air masses move away or are merged with each other in such a way that temperature and density variations are removed and there prevails a condition of almost uniformity in temperature and air density.

Conditions for Frontogenesis:

If the distribution of air fronts over the globe is closely observed, it appears that fronts originate only in limited areas which means that fronts originate only when some favourable conditions are available.

Temperature Difference:

- For frontogenesis to occur, **there must be a significant temperature difference between two opposing air masses.** One air mass should be cold, dry, and dense, while the other should be warm, moist, and light. When these two air masses converge, the colder, denser air mass invades the area of the warmer, lighter air mass, pushing it upward and forming a front.

> ***Do you know?***
>
> *Despite the convergence of two air masses at the equator, no front is formed due to the uniform temperature conditions of the trade winds. However, some meteorologists have described instances of frontogenesis occurring near the equator.*

Opposite Directions of Air Masses:

- Another essential condition for frontogenesis is the **convergence of two contrasting air masses.** When two thermally contrasting air masses meet face to face, they attempt to penetrate each other's regions, resulting in the formation of a wave-like front. In contrast, **when two air masses diverge, moving in opposite directions, this leads to the destruction of any existing fronts.** Patterson has identified four types of air circulation, with only the last two being conducive to frontogenesis.

Key Characteristics of Fronts:

Thickness of Frontal Zone:

- The thickness of a frontal zone is significantly **influenced by the temperature contrast** between the converging air masses. Generally, air masses with a **higher temperature difference result in a thinner, more distinct front** because they do not merge readily due to the pronounced differences in temperature.

Temperature and Pressure Changes:

- When there is a **sudden change in temperature across a front, it often correlates with a change in pressure.** This relationship between temperature and pressure shifts is a key characteristic of frontal zones.

Wind Direction Shift:

- A front typically experiences a **shift in wind direction due to the influence of the pressure gradient and the Coriolis force.** A Wind Shift is defined as a change in wind direction of 45 degrees or more occurring within 15 minutes, accompanied by sustained wind speeds of 10 knots or more throughout the wind shift.

Cloudiness and Precipitation:

- Frontal activity often leads to cloudiness and precipitation due to the warm air's ascent. As **warm air rises, it cools adiabatically, causing condensation and subsequent rainfall.** The type of precipitation (rain, snow, etc.) depends on the specific temperature and moisture characteristics of the air masses involved.

Slanting Boundary:

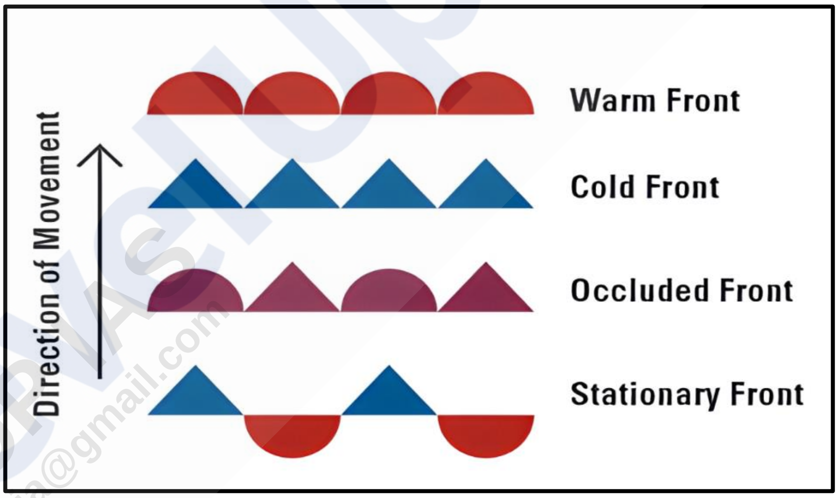
- **Fronts are not vertical walls but rather sloping boundaries,** typically tilted with a warm air mass overlying the colder air mass. This slant is due to the warm air mass being less dense and rising over the colder air.

Movement:

- Fronts are **not stationary; they typically move across the Earth's surface from west to east** due to the prevailing winds in the mid-latitudes. The speed of movement can vary depending on the pressure gradient and the properties of the air masses involved.

Classification of Fronts:

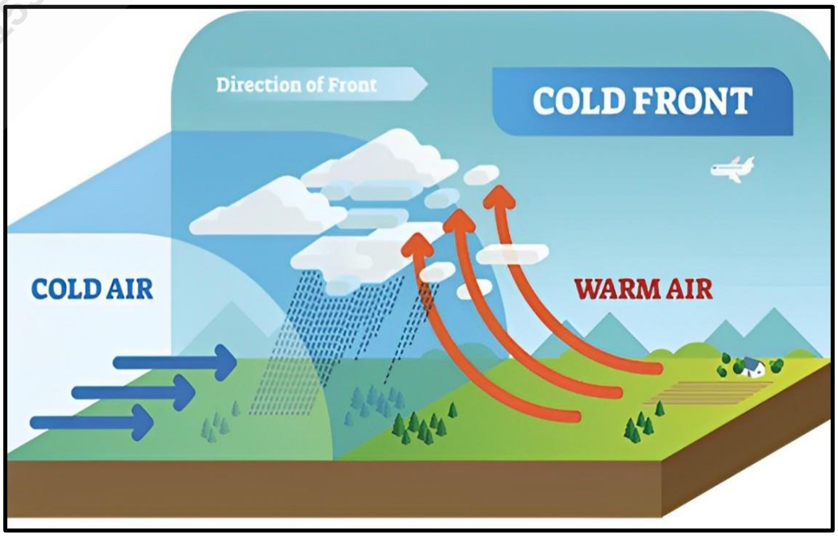
The identification of various types of fronts is based on observations of atmospheric conditions both at the surface and aloft.



There are four primary types of fronts:

- Cold Fronts
- Warm Fronts
- Occluded Fronts
- Stationary Fronts.

Cold Fronts:

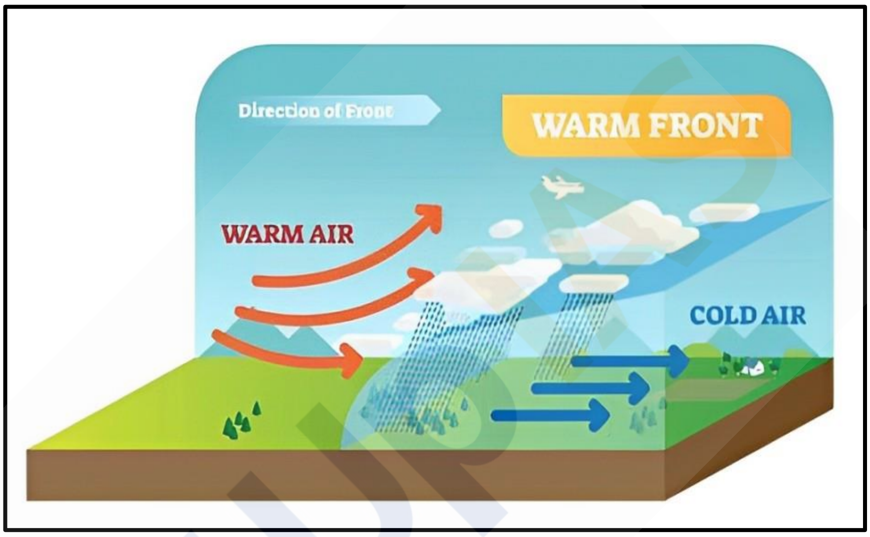


- A cold front is defined as the **boundary where cold air advances into a zone of warm air, forcing the warm air to rise due to the cold air's higher density.** This results in a **steep, rapidly moving transition zone.**
- Cold fronts generally **move quickly,** leading to **rapid changes in weather** conditions. They are characterized by a narrow band of cloudiness and precipitation, and their passage is marked by a **sudden drop in temperature** and a **wind shift** from the south to the west or northwest. The presence of friction can cause a squall line, leading to heavy showers.

Weather patterns associated with Cold Fronts:

- As a cold front approaches, there is an increase in wind within the warm sector, and the sky shows a sequence of clouds: **cirrus and cirrostratus** followed by **altocumulus and altostratus.** At the front, **nimbostratus and cumulonimbus clouds** produce heavy showers.
- **Precipitation** can occur ahead of or behind the front, depending on the air masses' physical characteristics. When cold air moves over warm water, it can lead to **heavy rain or snowfall.**
- With the passage of the cold front, the weather clears rapidly, with a significant **drop in temperature and humidity.** In winter, the passage of a cold front is often followed by a **cold wave,** further lowering temperatures.

Warm Fronts:

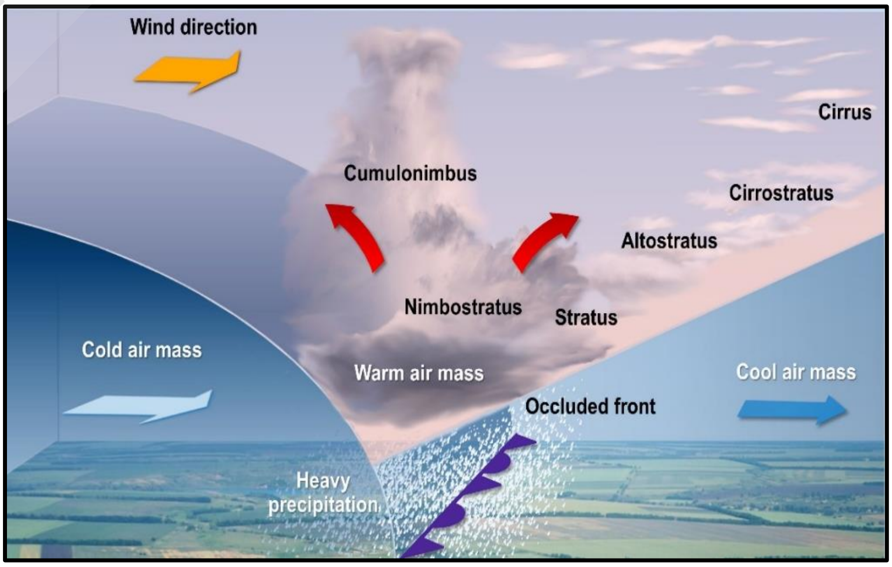


- A warm front is defined as a **gently sloping frontal surface where warm air actively moves over cold air, gradually occupying the territory formerly covered by cooler air.**
- As the warm air ascends the cold air wedge, it **cools adiabatically, leading to cloudy condensation and precipitation.**
- Unlike cold fronts, **temperature and wind direction changes are gradual.** Warm fronts yield **moderate to gentle precipitation over larger areas for several hours,** and convective activity is generally absent.

Weather patterns associated with Warm Fronts:

- The sequence of clouds preceding a warm front begins with **high-altitude cirrus clouds,** signalling the approach of the warm front. **Cirrostratus clouds,** forming ahead of the front, **produce halos around the sun and moon.**
- As the front approaches, clouds become lower and thicker, transitioning from cirrus, cirrostratus, altostratus, stratus, and nimbostratus to nimbus, with **steady precipitation extending over a long distance ahead of the front. Occasionally, cumulonimbus clouds and thunderstorms occur, though this is rare.**
- **During winter, temperature inversion** along the warm front can cause **freezing rain or sleet.** The passage of a warm front results in a **rise in temperature and pressure,** a small wind direction shift, increased specific humidity, and a change in weather conditions.

Occluded Fronts:



- An occluded front is defined as a front formed **when a cold front overtakes a warm front.** The **cold front moves more rapidly** than the warm front with the result that the **warm sector is progressively reduced in size.**
- **Ultimately the cold front overtakes the warm front and completely displaces the warm air at the ground.** Ultimately the cold and warm fronts

combine into one. Thus, a long and backward swinging occluded front comes into existence. **There are two types of occlusion:**

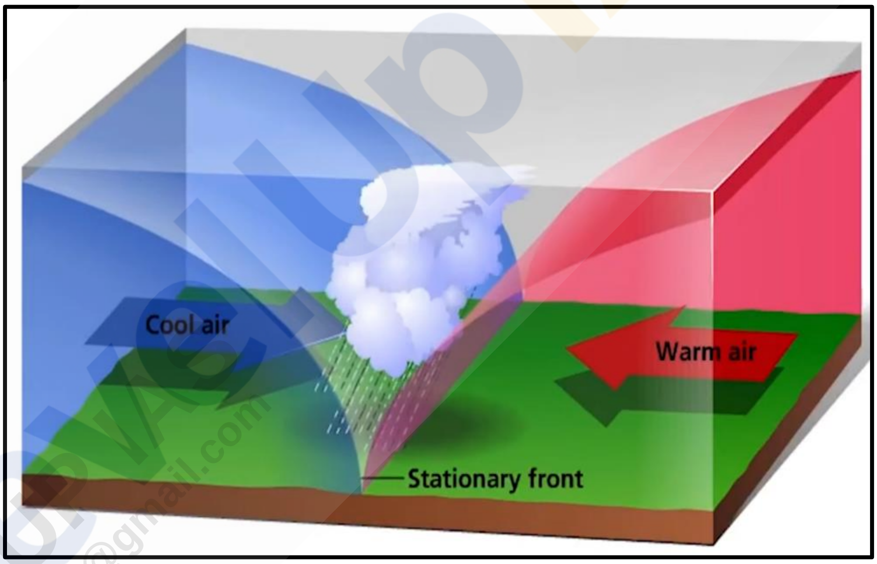
- **Cold Front Occlusion -** Occurs when the overtaking cold air is colder than the retreating cold air. Initially, weather resembles that of a warm front, but later transitions to conditions similar to a cold front.
- **Warm Front Occlusion -** Happens when the retreating cold air is colder than the advancing cold air. The less dense advancing cold air overrides the colder retreating air, often due to radiative cooling of the retreating air and the nature of the advancing air mass.

Weather patterns associated with Occluded Fronts:

- During a **cold front occlusion,** the **initial weather resembles that of a warm front, but as the occlusion progresses, it takes on the characteristics of a cold front.**
- In a **warm front occlusion,** the advancing cold air overrides the retreating, colder air, often due to the latter cooling by radiation. The uplift of the warm air mass in both types of occlusion leads to **cooling, condensation, and precipitation.**
- Thus, the weather at an occluded front typically **combines elements of both cold front and warm front weather,** leading to varied cloud formations and precipitation patterns.

Stationary Fronts:

- There are situations in which the surface position of a **front does not move** (when two air masses are unable to push against each other). Therefore, such a front is called a **stationary front.**
- The wind motion on either side of such a boundary is nearly parallel to the position of the front. Whenever some overrunning of warm air occurs along a stationary front, warm front type precipitation is likely to be produced.



Weather patterns associated with Stationary Fronts:

- **Cumulonimbus clouds** are formed. Overrunning of warm air along such a front causes **frontal precipitation.**
- Cyclones migrating along a stationary front can dump **heavy amounts of precipitation,** resulting in significant **flooding along the front.**

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